Physics — Grade 7 Formula Sheet

SI Base Units

Quantity	Unit	Symbol
length / distance	meter	m
mass	kilogram	kg
time	second	s
electric current	ampere	A
temperature	kelvin	K
amount of substance	mole	mol
luminous intensity	candela	cd

Metric Prefixes

Power	Prefix	Power	Prefix
10^{12}	T tera	10^{-1}	d deci
10^{9}	G giga	10^{-2}	c centi
10^{6}	M mega	10^{-3}	m milli
10^{3}	k kilo	10^{-6}	μ micro
10^{2}	h hecto	10^{-9}	n nano
10^{1}	da deka	10^{-12}	p pico

- No prefix $= 10^{0} = 1$ (the base unit).

Scientific Notation

$$a \times 10^{n}$$

- **Big** number → decimal moves **left** → **positive** n.
- **Small** number → decimal moves **right** → **negative** n.
- A prefix is just a power of ten: kilo $= 10^{3}$, nano $= 10^{-9}$.

$9{,}300{,}000 = 9.3 \times 10^{6}$
$0.0000045 = 4.5 \times 10^{-6}$

Unit Conversion

Multiply by fractions equal to 1; cancel unwanted units.

$$5\ \cancel{\mathrm{km}}\left(\frac{1000\ \mathrm{m}}{1\ \cancel{\mathrm{km}}}\right) = 5000\ \mathrm{m}$$

Speed units

$$\frac{\mathrm{km}}{\mathrm{h}} \div 3.6 = \frac{\mathrm{m}}{\mathrm{s}} \qquad \frac{\mathrm{m}}{\mathrm{s}} \times 3.6 = \frac{\mathrm{km}}{\mathrm{h}}$$

$$40\ \tfrac{\mathrm{km}}{\mathrm{h}} = 40 \times \frac{1000}{3600} = 11.1\ \tfrac{\mathrm{m}}{\mathrm{s}}$$

Common: $1\ \mathrm{km} = 1000\ \mathrm{m} \cdot 1\ \mathrm{h} = 3600\ \mathrm{s} \cdot 1\ \mathrm{min} = 60\ \mathrm{s} \cdot 1\ \mathrm{kg} = 1000\ \mathrm{g}$

Derived & Special Units

Quantity	Formula	Base units	Name
Force	$F = ma$	$\mathrm{kg \cdot m/s^2}$	Newton (N)
Energy / Work	$W = Fd$	$\mathrm{kg \cdot m^2/s^2}$	Joule (J)
Power	$P = W/t$	$\mathrm{kg \cdot m^2/s^3}$	Watt (W)
Frequency	$f = 1/T$	$\mathrm{1/s}$	Hertz (Hz)
Charge	$Q = It$	$\mathrm{A \cdot s}$	Coulomb (C)

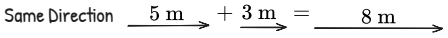
Symbols & Definitions

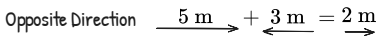
Quantity	Symbol	Unit	S/V
position	x	m	V
distance	d	m	S
displacement	Δx	m	V
time	t	s	S
speed	s	$\mathrm{m/s}$	S
velocity	v	$\mathrm{m/s}$	V
acceleration	a	$\mathrm{m/s^2}$	V
mass	m	kg	S
force	F	N	V
net force	F_{net}	N	V
weight	W	N	V
friction	f	N	V
normal force	N	N	V
tension	T	N	V
coeff. of friction	μ	—	S
moment	M	$\mathrm{N \cdot m}$	V
gravity	g	$9.8\ \mathrm{m/s^2}$	V
grav. constant	G	$6.67 \times 10^{-11}\ \mathrm{N \cdot m^2/kg^2}$	S

S = scalar, V = vector.

Scalars & Vectors

- **Scalar** — magnitude only (mass, time, distance, speed).
- **Vector** — magnitude **and** direction (position, displacement, velocity, acceleration, force).
- Collinear vectors (same axis): **same** direction → **add**; **opposite** → **subtract**.
- **Resultant velocity:** $v_{\mathrm{resultant}} = v_1 + v_2$ (on the same object).
- **Relative velocity:** $v_{BA} = v_B - v_A$ (velocity of B seen from A).





Direction	Sign
right / east / up / north	+
left / west / down / south	−

Kinematics

- Distance d = total path length (scalar). Displacement Δx = change in position (vector).
- $\Delta x = x - x_0$ (x final, x_0 initial position)
- Speed: $s = \dfrac{d}{t}$
- Velocity: $v = \dfrac{\Delta x}{t}$
- Average speed $= \dfrac{d_{\text{total}}}{t_{\text{total}}}$

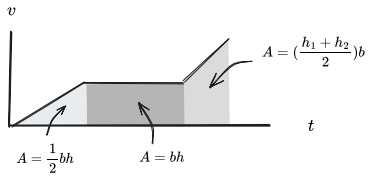
  Average velocity $= \dfrac{\Delta x_{\text{total}}}{t_{\text{total}}}$
- Acceleration: $a = \dfrac{\Delta v}{t} = \dfrac{v - v_0}{t}$
- Constant acceleration: $\bar{v} = \dfrac{v_0 + v}{2}$

Kinematic equations (constant acceleration)

$v = v_0 + at \quad \Delta x = v_0 t + \frac{1}{2}at^2 \quad v^2 = v_0^2 + 2a\Delta x$

Graph cheat sheet

Position–time: slope = velocity. **Velocity–time:** slope = acceleration; **area under the line** = displacement (above axis +, below −, unit m).



Forces

- **Newton's 1st law** (inertia): an object keeps its motion unless a **net force** acts.
- **Newton's 2nd law:** $F_{\text{net}} = ma$ (acceleration is in the direction of the net force).
- **Newton's 3rd law:** equal and opposite forces act on the **two** objects; they do **not** cancel.
- **Net force:** $F_{\text{net}} = \sum F$ (add same direction, subtract opposite).
- **Unknown force:** $F_{\text{unknown}} = F_{\text{net}} - F_{\text{others}}$
- **Weight:** $W = mg$, with $g = 9.8\ \text{m/s}^2$ on Earth.
- **Equilibrium:** $F_{\text{net}} = 0 \Rightarrow a = 0 \Rightarrow$ constant velocity. Static = at rest; dynamic = constant velocity.
- **Drag** increases with speed; at **terminal velocity** drag = weight, so $F_{\text{net}} = 0$.

$$1\ \text{N} = 1\ \text{kg} \cdot \frac{\text{m}}{\text{s}^2}$$

Friction

$$f = \mu N$$

- f = friction (N), μ = coefficient of friction (no units), N = normal force (N).
- **Kinetic friction** acts **opposite** the motion; **static friction** holds a still object in place.
- Bigger μ or bigger $N \rightarrow$ bigger friction.

Balancing Weight

When a single tension or normal force balances weight:

$$T = W = mg \qquad N = W = mg$$

Moments

- $M = Fd$ — M = moment $(\text{N} \cdot \text{m})$, F = force (N), d = **perpendicular** distance from the pivot (m).
- **Principle of moments:** a beam balances when $\sum M_{\text{clockwise}} = \sum M_{\text{anticlockwise}}$.

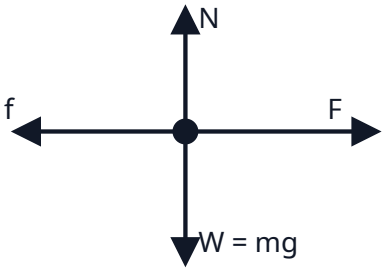
Universal Gravitation

$$F = G\frac{m_1 m_2}{r^2}$$

- $G = 6.67 \times 10^{-11}\ \text{N} \cdot \text{m}^2/\text{kg}^2$. Force grows with the masses, shrinks with the square of the distance.

Free-Body Diagrams

Draw every force acting on **one** object as a labeled arrow from a dot. Include only forces **on** the object; arrow length = magnitude, head = direction.



Problem-Solving Steps

1. Annotate: what is given? what is the unknown?
2. What equation connects the quantities?
3. Plug in with units and solve.
4. Check the units and direction.

Hints

"Starts from rest" → $v_0 = 0$. "Comes to rest" → $v = 0$. Give a direction for every vector quantity.